

This workbook has both a paper and a Jupyter notebook submission.

1. Exercise 5.15
2. Exercise 5.16
3. Create a graph of the total error (approximation + precision) of the derivative as a function of h similar to what is shown on the “Derivatives_1_notebook” for the function

$$f(x) = -3x^3 + 4x^2 + 6x - 1 \tag{1}$$

at $x = 1.5$. Compare 3 different central methods:

$$\frac{df}{dx} \approx \frac{f(x + h/2) - f(x - h/2)}{2} \tag{2}$$

$$\frac{df}{dx} \approx \frac{\frac{1}{24}f(x - 3h/2) - \frac{27}{24}f(x - h/2) + \frac{27}{24}f(x + h/2) - \frac{1}{24}f(x + 3h/2)}{h} \tag{3}$$

and

$$\frac{df}{dx} \approx \frac{-\frac{3}{640}f(x - 5h/2) + \frac{25}{384}f(x - 3h/2) - \frac{75}{64}f(x - h/2) + \frac{75}{64}f(x + h/2) - \frac{25}{384}f(x + 3h/2) + \frac{3}{640}f(x + 5h/2)}{h} \tag{4}$$

Comment on the differences.